

# Lei Xu

## CONTACT

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## PROFILIO

I am a dynamic researcher with expertise in deep learning, machine learning, and computer vision areas. Proven ability to lead innovative projects and develop advanced algorithms. Strong programming skills complemented by excellent communication and teamwork abilities. Quickly adapt to new tools and frameworks, such as RAG, Agent/Skills, etc, to enhance project outcomes and personal expertise.

## PROFESSIONAL EXPERIENCE

- Postdoctoral Researcher | 2026–Now**  
Tampere University
  - Conducted ECG representation learning with generative models and a graph concept network, leading to one paper.
  - Working on the “Geometric AI” project, which has led to two published papers on image restoration/deblurring to date.
  - Working on Quasi-signal representation learning, which will lead to a paper.
- Postdoctoral Researcher | 2024–2025**  
PoDoCo-grant
  - Working on the “GreenVISA: Energy-Efficient VISual Analysis for Industry-Scale Machine Vision Services” project, leading to two published papers.
- External Researcher | 2022–2023**  
Nokia Technologies Ltd.
  - Working on the VCM project, which aims to explore deep learning-based algorithms for video compression
- Researcher Scientist | 2022–2023**  
Silo AI
  - Leading research on the deployment of computer vision tasks on the Azure platform and hands-on research of AI models development on Nvidia Jetson platforms based on DeepStream SDK.
- Doctoral Researcher | 2017–2024**  
Tampere University
  - Working on solving multi-class imbalanced problems with possibility models
  - Working on solving imbalanced problems existing semantic segmentation with generative models
- Researcher | 2016–2017**  
Tampere University
  - Working on solving imbalanced problems with LDA

## EDUCATION

- Tampere University | 2018–2024**  
Doctor of Science (Technology)
- Tampere University | 2014–2017**  
Master of Science (Information and Technology)
- East China Normal University | 2002–2006**  
Bachelor of Science (Communications Engineering)

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## SKILLS

- Competitive skills in deep learning frameworks: Keras, TensorFlow, PyTorch, NumPy, Hugging Face Transformers, RAG, etc.
  - Competitive skills to solve real-world tasks with deep learning algorithms.
  - Good skills in code control tools: GIT and Bitbucket.
  - Strong programming skills in Python, Matlab<sup>TM</sup>, GPU cluster, and Linux shell scripting, etc.
  - Good understanding of Azure IoT Edge platform, Claude, and DeepStream SDK.
  - Strong skills in English, written and verbal communication.
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## PUBLICATIONS

- Lei Xu, Fahad Sohrab, Mehmet Yamac, Merja Heinaniemi, Moncef Gabbouj, "From Reports to Ontologies: Ontology-Guided Representation Learning for 12-Lead ECG." arXiv preprint arXiv:2605.25883 (2026).
- Yamac, Mehmet, Lei Xu, "EMTS-MHGNet: Efficient Real-World Image Deblurring via Multi-Scale Tensorial Feature Extraction and Lightweight Gated Reconstruction." CVPR Workshop (2026).
- Yamac, Mehmet, Lei Xu, Serkan Kiranyaz, Moncef Gabbouj, "MTS-CSNet: Multiscale Tensor Factorization for Deep Compressive Sensing on RGB Images." ICIP (2026).
- Lei Xu, Mehmet Yamac, Mete Ahishali, and Moncef Gabbouj. "Multi-Scale Tensorial Summation and Dimensional Reduction Guided Neural Network for Edge Detection." In Proceedings of the IEEE/CVF International Conference on Computer Vision. 2025.
- Lei Xu, Mete Ahishali, and Moncef Gabbouj. "Dropout concrete autoencoder for band selection on hyperspectral image scenes." IEEE Geoscience and Remote Sensing Letters (2025).
- Liang, Aoming, Qi Liu, Lei Xu, Fahad Sohrab, Weicheng Cui, Changhui Song, and Moncef Gabbouj. "Conformal Prediction on Quantifying Uncertainty of Dynamic Systems." arXiv preprint arXiv:2412.10459 (2024).
- Lei Xu and Moncef Gabbouj. "Revisiting generative adversarial networks for binary semantic segmentation on imbalanced datasets." arXiv preprint arXiv:2402.02245 (2024).
- Lei Xu, Jenni Raitoharju, Alexandros Iosifidis, and Moncef Gabbouj. "Saliency-based multilabel linear discriminant analysis." IEEE transactions on cybernetics 52, no. 10 (2021): 10200–10213.
- Lei Xu, Honglei Zhang, Jenni Raitoharju, and Moncef Gabbouj. "Unsupervised facial image de-occlusion with optimized deep generative models." In 2018 Eighth International Conference on Image Processing Theory, Tools and Applications (IPTA), pp. 1–6. IEEE, 2018.
- Lei Xu, Alexandros Iosifidis, and Moneef Gabbouj. "Weighted linear discriminant analysis based on class saliency information." In 2018, the 25th IEEE International Conference on Image Processing (ICIP), pp. 2306–2310. IEEE, 2018.
- Huttunen, Heikki, Ke Chen, Abhishek Thakur, Artus Krohn-Grimberghe, Oguzhan Gencoglu, Xingyang Ni, Mohammed Al-Musawi, Lei Xu, and Hendrik Jacob Van Veen. "Computer vision for head pose estimation: review of a competition." In Scandinavian conference on image analysis, pp. 65–75. Cham: Springer International Publishing, 2015.